

Statistical Identification of a Distance Metric that Maximizes Interrater Reliability

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Abstract

In this paper, we investigate 3 different distance metrics and determine which of the 3 has the highest Inter-Rater Reliability.

1 Introduction

The quality of a lesson is directly related to the quality of the participation and interactions of students with the teacher, content, as well as other students. Melhuish et al [4] has defined a rubric for classifying different possible interactions in 33 different ways, and then assigning overall quality ratings to the lessons.

However, given the large number of codes and the moment by moment nature of the data recorded, it is unrealistic that two observers of the same lesson would end up with the exact same list of codes and time stamps. For instance, it is possible that two different people may view the same interaction as different classifications, or perceive the interactions to begin at different times in the lesson. For this reason, in order for an observation tool to be useful, it is crucial to assess the interrater reliability, i.e to measure the typical level of agreement between multiple raters or judges. Given the complex nature of the Math Habits Tool and classification scheme, measuring its interrater reliability (IRR) is not straightforward and requires a novel distance metric in order to capture the key differences between the classifications of two different viewers of the same lesson and to align with the goal of capturing moments during the lesson that are mathematically productive in similar ways and are seen by two different people to occur at similar moments throughout the lesson.

2 Terminology

3 Research Question

The research question we will examine is as follows: how can we determine a reliable distance metric to quantify the inter-rater reliability between coders who have coded subjective qualitative data on the quality of students' and teachers' interactions with each other and with the content of different mathematics lessons? In addition, how can we predict the overall Teacher Code from the norm of the lesson?

4 Preliminary Literature Review

A preliminary literature review has produced previous work based off of the Math Habits of Mind Framework of student-student, student-teacher, and student-content interactions, which have governed the way in which coders assess and encode the different actions conducted by students and teachers during a lesson [1, 2, 3]. Based off of this data, the relationship between coder-assigned Math Habits of Mind action codes and the overall classroom codes for students and teachers has been identified with over 70% accuracy [4]. In addition, the data metrics of "total" and "spread", as defined by Melhuish et al have been used to identify which action codes have the greatest effect on the overall lesson quality.

However, previous literature lacks the consideration of how we can educate teachers and coders in such a manner to help them identify a lesson to effectively teach mathematical thinking to the students. In other words, whether two different coders can produce reliable results. For this reason, we wish to produce a novel distance metric that can quantitatively allow us to analyze whether the goal stated above is achievable. Therefore, we would be able to tell whether certain “trainings” for teachers would be effective in educating them on how to teach an optimal lesson, as viewed by the mathematics education department.

5 Objectives

The long-term goal is to develop a way or system to determine the quality of lessons.

The objective of the study is to provide a quantitative relationship of how different coders analyze a lesson by measuring the difference in the data between the two people, so we can determine the inter-rater reliability. This can also help to identify the importance of different activities of a classroom when coding the data.

Finding a reliable distance metric would help quantify the differences in the subjective classifications of math lessons by two different people, and serve to help determine whether trainings actually are effective in helping educators define a common perception of what a “good quality” mathematics lesson truly is.

6 Methodology

1. Coding

To measure the quality of the teachers and interactions, Melhuish et al. [6] developed the Math Habits Tool which captures mathematically productive components of classrooms in terms of both student and teacher in-the-moment actions. When applied to a lesson, this tool provides a rich time-stamped data set listing the observable teaching routines, catalytic teaching habits, student habits of mind and habits of interaction. This was done by recording various different classes imparted within different districts. Afterwards, a group of coders assigned every action done within the classroom to one of the 33 classifications that describe the type of interaction.

2. Data Reduction

- **Splitting Intervals** In order to reduce the amount of data that we will need to analyze, we start by splitting the lessons into ten equivalent intervals. Given that we have 33 possible codes for an action, this gives us a total of 330 data points to analyze.
- **Total and Spread** We utilize two different data metrics, which are the “total” and the “spread”. The “total” counts the number of times that a specific code appears during the lesson, whilst the “spread” measures how spread out said code is during the lesson.
- **Collapsible Data** To reduce the amount of data we need to analyze, we can combine similar codes into a single code (NewCode). After collapsing the data, a new data set is created, which we can also use for analyzing.

3. Distance

We have tested out various different distance metrics in order to see which one is the best fit for the data that we are using.

4. Assessing Reliability

To alter our data to our most desired form, we must use different methods/equations to find the difference between two data points. For example, the methods of Euclidean, Canberra, and Manhattan each have a different algebraic equation that calculates the distance between two data points in different ways. The equations of the above methods are as follows [5]:

Euclidean:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\left(\sum_{i=1}^n (x_i - y_i)^2 \right)}$$

Canberra:

$$C(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n \frac{|X_i - Y_i|}{|X_i| + |Y_i|}$$

Manhattan:

$$M(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|$$

Using all three of these methods, we will try to produce a desirable outcome by finding the best way to measure the distance in a quantitative way. A desirable outcome will be defined as the variance being as close to 1 as possible. The **variance** is defined as:

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

where x is a single element of the differences, μ is the mean of the total differences, and N is the total count of the differences. The range from 0.7 - 1 is what we determine as “desirable”. On the graph, these numbers translate to the “FALSE” (red) data points being concentrated to the right, and the “TRUE” (blue) data points should be concentrated to the left. Additionally, we are planning on using both the “Monte Carlo Method”, as well as “Bootstrapping” to assess the reliability of our model. Both of them consist of estimating numerical values based on small data samples.

7 Results

After numerous trials, with several distance metrics, we have obtained some preliminary results that give us a very good idea of how reliable our model is.

The red/FALSE part of the graphs show the difference between two coders coding different lessons. The blue/TRUE part of the graphs show the difference between two coders coding the same lesson. The x-axis represents what the quantitative difference between two selected data points are. The y-axis shows the count of each difference between two coders. Our ideal situation would be that the red/FALSE part of the graphs are skewed left because the difference should be greater since the two coders are coding different lessons, while the blue/TRUE part of the graphs are skewed right because the difference should be minimal since the coders are coding the same lesson. There are 31 data points for the blue/TRUE part because there are 31 different lessons that two different coders coded. There are $\binom{62}{2} - 31$ data points for the red/FALSE part because there are 62 different lessons, and we have to subtract the possible cases of choosing the same lessons coded by different coders.

The “**Euclidean Method**” is used to find the shortest distance between two points by using the square root function in a way similar to the “Pythagorean Theorem”. It is useful if you’re working in a continuous space where all dimensions are properly scaled and relevant.

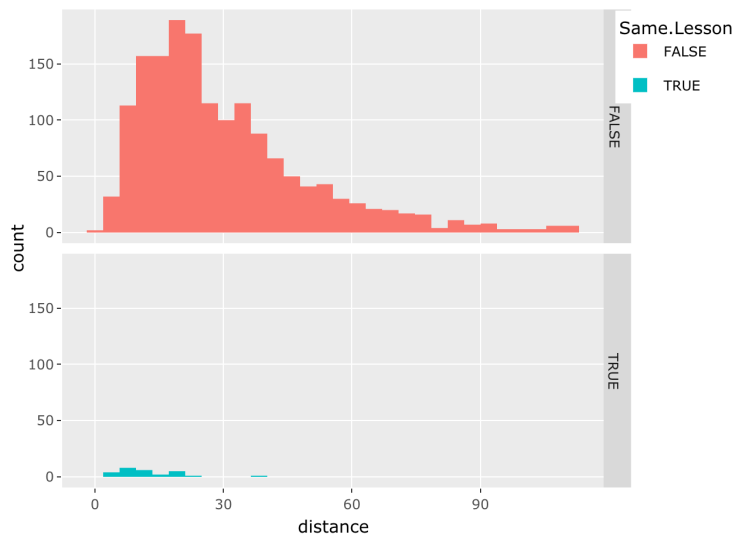


Figure 1: Lesson Count vs. Distance using the “Euclidean Method” with the collapsed variables. Variance = 0.857711

Even though the red/FALSE part of the graph is skewed right instead of left, the blue/TRUE part of the graph is optimal for what we want. This results in the variance being in the range that we want it to be in.

The “**Canberra Method**” uses absolute values to calculate the distance between two points, and is usually used as a list for comparing ranked list, which is probably why it gave us such a low variance when tested.

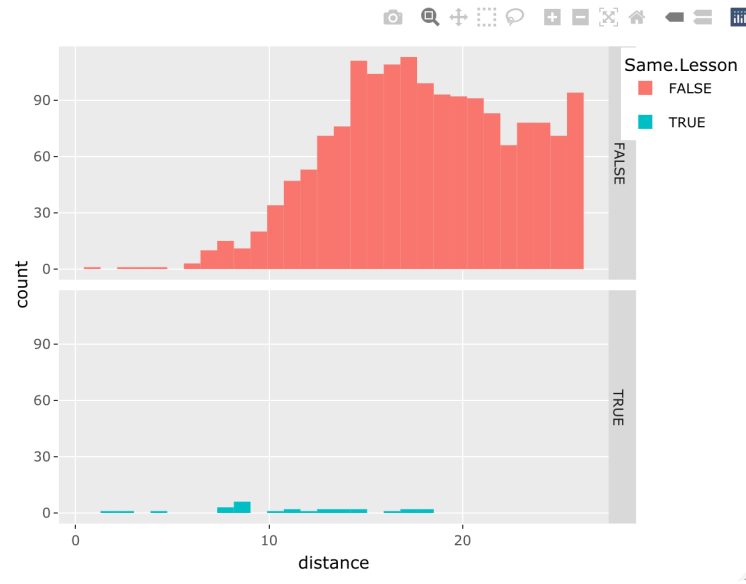


Figure 2: Lesson Count vs. Distance using the “Canberra Method” using the spread variables. Variance = 0.1137029

Even though the red/FALSE part of the graph is skewed left, which is ideal, the blue/TRUE part of the graph is not skewed right and is spread throughout the x-axis. This results in our variance being extremely low and far away from 1, so the Canberra method is not ideal for our situation.

The “**Manhattan Method**” is based on using the absolute value, as opposed to using the squared error, as the “Euclidean Method” does. As such, the results from the two are very similar, but the absolute value tends to give us more robust results, whilst the “Euclidean Method” could be influenced more by outliers, as it tends to exaggerate the discrepancies. You can imagine the Manhattan Method as going from one data point to another by blocks, like a Manhattan street plan.

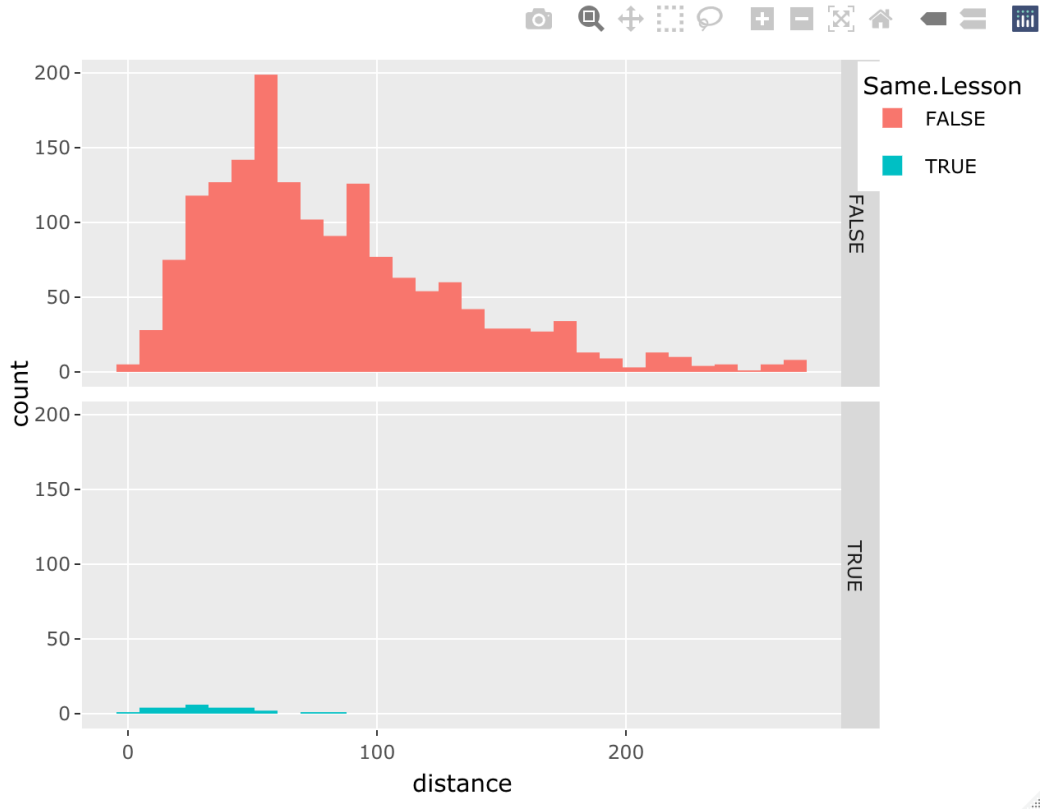


Figure 3: Lesson Count vs. Distance using the “Manhattan Method” using the spread variables. Variance = 0.854242

Even though the red/FALSE part of the graph is skewed right instead of left, the blue/TRUE part of the graph is optimal for what we want. This results in the variance being in the range that we want it to be in.

References

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